

Real-World Presence in the Era of Ubiquitous Computing

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ABSTRACT

In the era of everyday ubiquitous computing, all of us have at least once lost track of time, space or our social environment when using our smartphone. In such situations, we might not feel present in the here and now in the physical real world. Presence has been a long term topic of interest in virtual reality research as the feeling of being there, in a technology-mediated virtual world. However, we have yet to understand our feeling of being here and now in the physical real world under the wide availability and easy accessibility of smartphones. In my thesis, I want to examine the concept of real-world presence from a theoretical, users' and technology development perspective. I aim at understanding current and developing new mobile solutions, with the goal of supporting our presence in the physical real world.

KEYWORDS

Presence, Ubiquitous Computing, Everyday Computing, Mindfulness, Technology Non-Use, Technology Disengagement, Empirical Studies, Literature Review

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1 INTRODUCTION

Probably everyone has at least once lost track of their immediate physical or social surroundings while interacting with their smartphone. For example, we might have briefly halted a conversation when a smartphone notification popped up. Or we aimed at taking that one perfect shot over and over again for social media, losing the moment of awe when visiting a new place. On the other hand, we might have video-called family and friends from around the world, feeling to be together at one place. In situations such as these, we disengage to some extent from the *real world* and immerse into a *digital world*. However, people report fearing disengagement from the real world when interacting with technology [1]. Some state that it causes stress and negative emotions [9]. Various public and academic sources, including (ironically) social media, raise voices of concern over smartphones' impact on our social and personal life (e.g., [3, 5]).

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We could argue that the solution is quite obvious and simple: to leave smartphones at home and go get a walk in the park. However, smartphones have become so intertwined with our personal and professional lives. With smartphones, we can connect each other, listen to music and watch movies, even do our work. This *one device* is a widely available and easily accessible portal to a virtual *digital world*. As if we live in parallel worlds, the real and a digital world, continually moving back and forth. To mitigate the disturbing feeling of disengagement, we will need technology solutions in the future that will support our presence across these parallel worlds.

Waterworth [15] says that "*presence is about the present, the here and now in the physical and virtual world.*" The concept of presence is well known in research works on virtual reality (VR). It describes either the perceived realness of the VR or the subjective feeling of *being there* in the VR [10]. Such presence is frequently measured by the extent of user's unawareness of the real world: the higher the unawareness, the better the presence (in VR) [16].

Yet, in my thesis, I want to understand our presence in the *real, physical world*, that is, our feeling of *being here*. My interests in this topic evolve not only around our active interaction with a virtual digital world, but also the way real-world presence unfolds under the mere existence of a ubiquitous digital world accessible at our fingertips. This leads to the first research question of my thesis:

RQ1: "When we are there (i.e., in the digital world), to what extent are we still here (i.e., in the real world)?"

Currently, we simply track screen time and the number of times we unlocked our mobile device to measure disengagement from the real world. However, I am interested in finding 1) measurements for real-world presence and 2) the optimal balance between presence in the real and presence in the digital world in everyday computing. Once I have established an understanding of real-world presence and its influencing factors, I want to develop and evaluate new solutions that support our knowledge of the real world and the feeling of being in the real world, summed up as:

RQ2: "What technology designs can support our real-world presence?"

We yet have to investigate ways of anchoring the user in the real world, the *here* and *now*, when a digital world is at hand. Whether and to what extent technology can help achieving that goal is the question of interest.

2 BACKGROUND

The feeling of presence has been studied by scholars across various disciplines ([7, 12] provide extensive reviews on the used definitions and conceptualizations of presence). The leading research society on presence in computer science, the *International Society for Presence Research*, considers "*presence, short for telepresence, [to happen] when people use technology and overlook at least part of its*

role in the experience.” [10]. Lombard and Ditton [6] call this presence conceptualization *presence as realism*. In other words, when designing a VR world, designers lean onto the real world. The real world is the ideal to strive for, it is the unquestionable truth.

Opposed to mimicking reality, Waterworth et al. [15] state that “*presence is about the present, the here and now in the physical and virtual world.*” This aligns with the *transportation* idea of presence [6]: technology is conveying us into another world and inducing a feeling of “being there” in the technology-mediated world. Most prominently, these feelings have been researched in the context of virtual reality (VR). In current VR systems, the user puts a head-mounted display (HMD) on and gets immersed with as much senses as possible into the virtual world. The optimal HMD experience excludes the user’s awareness of the real world, i.e., the user should feel a maximal presence in the virtual world [16].

However, even traditional media can absorb us within their world too, providing a feeling of being in an alternative reality. Oral storytelling and books are early examples of such “technology”. Whereas the feeling of presence, that is, the feeling of being there in VR has been thoroughly investigated [12], and to some extent in other technology-mediated worlds (e.g., television [4]), our knowledge on the feeling of *being here* in the real world has been relatively sparse. Gergen [2] elaborates on the challenges of an *absent presence*, an individual “*physically present, but [...] absorbed by a technologically mediated world elsewhere.*” The same author describes that radio, film, television, as well as notebooks, smartphones and the internet, are all able to create an absent presence. I plan to build on that work by examining the extent of real-world presence in the absent presence created by a smartphone. In the following chapter, I describe my research approach on closing the gap.

3 RESEARCH APPROACH

To answer my research questions, I will examine the concept of real-world presence from different perspectives:

- (1) **theoretical/conceptual** perspective: the way real-world presence can be conceptualised and operationalised
- (2) **users’** perspective: the way users manage their presence across the real and digital world
- (3) **technology design** perspective: the way technology can be designed to support users in strengthening their real-world presence by the means of technology itself

Primarily, I want to understand and evaluate the different dimensions of *real-world presence*. As result, I want to establish a conceptualization of real-world presence in two circumstances: 1) under the mere existence of and 2) during active interaction with smartphones. Although similar, I differentiate the two circumstances, as one might feel less present even when not actively engaging with technology, but knowing it is there, and “waiting” for user’s attention at any time. Following, I want to review and analyze existing literature on the topic of presence and its counterbalance immersion with the goal of transferring current knowledge on presence in a technology-mediated world to presence in the real world.

The second perspective evolves around understanding users - their perception, feelings and mental models - when tangled between different worlds. A set of qualitative methods (e.g., surveys,

interviews, diary studies and focus groups) will examine users’ perception of being here and now in the real world while engaging with mobile devices and services and 2) the strategies users employ to switch between the worlds.

Examining the third perspective includes the design and empirical evaluation of technology solutions that reinforce real-world presence in the field. The first task is to explore and evaluate designs that can break us free, temporarily or permanently, from mobile devices and/or parts of their services. However, the more challenging task is to find remedy between the intuitively contradicting real-world presence and technology use. Tackling this challenge should unfold in several complementing steps. First, I want to find meaningful application contexts and user groups and second, conduct interactive design sessions with both non-expert and expert users. The goal is to develop new designs to be empirically evaluated in the field and by experts.

In the following two chapters, I present results to date and a research agenda in form of projects I currently work on. This list is not exhaustive nor final, but presents a rather rough research plan, leaning onto the described approach above.

4 RESULTS AND CONTRIBUTION TO DATE

Analysis of Mindfulness Research in HCI. My main contribution so far has been reviewing and analyzing the HCI research corpus on the usage of the term “mindfulness”, published in CHI19 [13]. Mindfulness is strongly tied to presence. The most used definition of mindfulness in HCI denotes that mindfulness is “*the awareness that emerges through paying attention on purpose, in the **present moment**, and non-judgmentally to the unfolding of experience moment by moment.*” As such, the project helps in understanding real-world presence in the first place (RQ1).

Besides definitions, the review consolidates the different conceptualizations, practices, technologies, evaluations and scientific findings that have been developed around mindfulness in HCI research. The analysis yielded a framework for mindfulness consisting of four perspectives on mindfulness, along with seven lines of research, which represent common research motivations and topics:

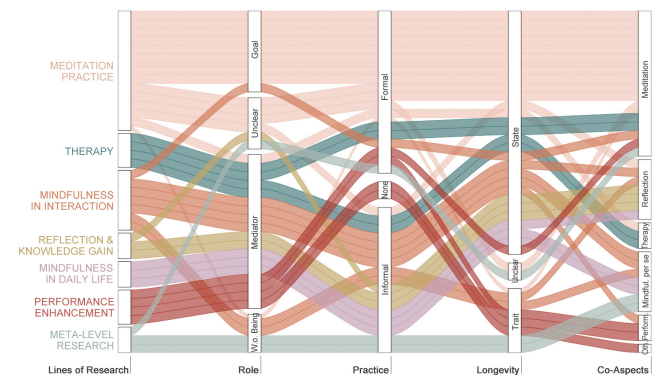


Figure 1: An alluvial diagram of the four perspectives on mindfulness and seven lines of research resulting from a systematic literature review on the usage of “mindfulness” in HCI [13]

Research Project	Research Question	Status
Analysis of Mindfulness Research in HCI	RQ1: Understanding RWP & RQ2: Design of Technology	Completed: Published
The User Between Worlds	RQ1: Understanding RWP	In Progress
Supporting Meditators with Apps	RQ2.1: Understanding Existing Technology to Support RWP	Completed: To be Submitted
A Disengagement Design Space	RQ2.1: Understanding Existing Technology to Support RWP	In Progress
Internet Memes for Self-Reflection	RQ2.2: Designing New Technology to Support RWP	Completed: To be Submitted
Irreversible Design	RQ2.2: Designing New Technology to Support RWP	In Progress

Table 1: A preliminary overview of the described research projects, the research question they tackle and their current status. The abbreviation RWP stands for real-world presence.

(A) *Meditation Practice*, (B) *Therapy*, (C) *Reflection & Knowledge Gain*, (D) *Mindfulness in Daily Life*, (E) *Mindfulness in Interaction*, and (F) *Performance Enhancement* (cf. 1). The interplay of mindfulness and technology use in everyday life, and thus real-world presence, comes particularly to prominence in the lines C, D and E. In C, practicing mindfulness means observing the self in the present, complementary to reflecting on the past and future. Technology acts as mediator in the observation. In D, mindfulness is the raised awareness in decision making in everyday contexts, with technology informing the decision making process. Finally, in E, the technology design itself stimulates a more direct involvement in interaction, leading to more engagement with and care for what a person is doing [14]. The lines of research lay ground for RQ2, as these can provide a frame for the development of new solutions.

Supporting Meditations with Mindfulness Apps. This project is to be submitted. It evolves around meditators' notion of mindfulness and experience using mindfulness meditation technology (i.e., apps) and, in particular, the routine around using these apps. A qualitative analysis of interviews ($n = 12$) and a subsequent design workshop ($n = 6$) add to the controversial discussion on the (interfering) role of technology in mindfulness meditation by drawing a differentiated, user-centered, picture.

Internet Memes for Self-Reflection. This project is to be submitted. We built and evaluated ($n = 30$) a mobile journaling app to examine the usage of internet memes for self-expression and self-reflection in daily journaling. We were interested in how an immersive content, such as internet memes, can be given a new, more meaningful, purpose. Our results showed that users have a sharpened focus on everyday happenings when journaling with internet memes.

5 NEXT STEPS

The following three projects are still in progress. An overview of all presented projects can be found in Table 1.

The User Between Worlds. To tackle RQ1, we are collecting and categorizing situations in which people felt to be between worlds: the real world, the technology-mediated world and our inner world. For example, our work colleague may pull us back to the real, physical world by knocking at the door, while we are listening to music on noise canceling headphones during a work task – or simply getting lost in the music (mind-wandering or daydreaming). There are many examples similar to this, as first results of a conducted focus group with meditators and presence researchers ($n = 6$) show.

Currently, we are gathering responses from a subsequent questionnaire on strategies people employ to manage their presence across the three worlds, resulting in a taxonomy of user strategies.

A Disengagement Design Space. With respect to RQ2, there have been recent calls for “undesigning” technology by technology design [11] and transferring mindfulness to everyday real life practices [8]. Digital detox is another recently emerged term, with the goal of rebalancing our digital consumption and our real-world presence. What these two concepts have in common is their call for purposefully designing technology we can *gradually* disengage from, opposed to restrictive means of enforcing non-use, e.g., limiting screen time. Thus, my project's goal is to deduce a design space from existing literature on designs made with disengagement as their goal. I plan to test a series of exemplary prototypes evolving from the design space.

Irreversible Design. The goal is to examine in what way irreversible actions can act as design strategy. We wonder in what way our perception of digital content changes, if we behave towards the digital as we behave when tearing paper or breaking glass, i.e. materials that cannot be restored by the means of an undo button. We expect to gather implications for the design of awareness-raising technology, and technology we care for.

6 LONG-TERM GOALS

My dissertation has three long-term goals, aligned with the presented perspectives. From the theoretical perspective, the goal is to coin the term real-world presence, with the hope of it becoming a measure for good technology design. On the technology design side, my results should shed more positive light onto smartphones, opposed to recent media headlines concerning their negative impact on our lives. Most importantly, I expect the outcomes to help *people* better balance their presence across the real and digital world.

7 HISTORY & TIMELINE

I have started my PhD mid 2017 at the LMU Munich, under the supervision of Heinrich Hussmann. Before that, I worked as an IT consultant in the industry, followed by a research and teaching assistant position at the TU Munich. While at the TUM, I investigated how mobile apps can support the decision-making process for healthy eating. I continued that research at LMU, but later on got more interested into self-reflection and mindfulness within HCI. This ultimately led me to the presented topic. After giving birth to my son in August 2019, I took a year parental leave. I plan to submit and defend my thesis in the second half of year 2022.

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